

ML Foundations — Day 4: Model Tuning, Regularization & Reproducible Pipelines

Dataset: Adult / Census Income (UCI, via OpenML) | Qasim | random_state = 42

Hyperparameter Search Summary

This environment has a single CPU core; a quick benchmark found Logistic Regression's **saga** solver with L1 penalty took ~80s for one fit vs. ~0.3s for **liblinear** on this dataset, so saga was dropped from the search (liblinear covers both L1 and L2 anyway) and RandomizedSearchCV used reduced n_iter/cv (documented in the notebook) to keep the search runnable end-to-end.

Model	Best CV ROC AUC	Search time	n_iter × cv	Best parameters
Logistic Regression	0.9111	70.4s	30 × 3	penalty=l1, C=0.541, solver=liblinear, max_iter=2000
HistGradientBoosting	0.9253	119.0s	15 × 3	learning_rate=0.028, max_iter=200, max_depth=None, l2_regularization=0.0054, min_samples_leaf=20

HistGradientBoosting remains the stronger model after tuning (0.9253 vs 0.9111 ROC AUC), consistent with Day 3, and was carried forward for diagnostics, calibration, and final evaluation.

Diagnostics

Learning curve (HistGradientBoosting, tuned): train ROC AUC 0.9405, validation ROC AUC 0.9253, gap 0.0151. This is a small, stable gap — not a strong overfitting signal, and both curves plateau well above the Day 1 baseline, so no underfitting either.

Validation curve — Logistic Regression C: the train/validation gap grows steadily with C (0.0006 at C=0.001 up to 0.0033 at C=100), and validation ROC AUC peaks at **C=1.0**. Below C=1 the model is slightly under-regularized-safe (small gap, but lower validation score); above it, the growing gap signals mild overfitting risk without a validation-score benefit — C=1 is the sensible operating point.

Validation curve — HistGradientBoosting min_samples_leaf: the gap shrinks monotonically as min_samples_leaf grows (0.009 at 20 down to 0.0062 at 400), confirming higher min_samples_leaf is stronger regularization as expected; the tuned value of 20 (from the search) sits at the higher-gap end, but the learning curve above shows this is still a small, acceptable gap in absolute terms, traded for the highest raw validation score.

Calibration & Threshold

Brier score before calibration: **0.0865**. After CalibratedClassifierCV(method='sigmoid', cv=5): **0.0862** — a small but real improvement (−0.0003), so calibration was **kept** per the improve-or-skip rule. The reliability diagram shows both curves already fairly close to the diagonal (HistGradientBoosting tends to produce reasonably well-calibrated probabilities out of the box), with sigmoid calibration nudging it slightly closer.

Threshold tuning (maximize F1, on the dev set): optimal threshold = **0.40** (precision 0.730, recall 0.746, F1 0.738 at that point) vs. the default 0.50 threshold (precision 0.797, recall 0.656, F1 0.720). Lowering the threshold to 0.40 trades some precision for a meaningful recall gain (+9 points), raising F1 by ~1.8 points — in the marketing-targeting context, this means the campaign catches more genuinely high-income households at the cost of a modestly higher false-positive (wasted outreach) rate.

Final Test Performance

Threshold	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
0.50 (default)	0.8760	0.7930	0.6523	0.7158	0.9279	0.8302
0.40 (tuned)	0.8692	0.7253	0.7297	0.7275	0.9279	0.8302

Evaluated once on the untouched hold-out test set (never used in search, calibration fitting, or threshold selection). Test ROC AUC (0.9279) sits essentially on top of the tuned CV ROC AUC (0.9253, well within one fold's spread) and even slightly above Day 3's untuned CV estimate (0.9262) — performance is stable from CV to test, with no sign the search overfit to the CV folds. ROC and Precision-Recall curves and the confusion matrix at the tuned threshold are in the notebook.

Production Readiness Note

The saved artifact (**final_pipeline.pkl**, ~4.5MB) is a single scikit-learn **CalibratedClassifierCV** wrapping the full tuned HistGradientBoosting pipeline (feature engineering → preprocessing → model), so it accepts a raw DataFrame with the

same columns as the original data and requires no separate preprocessing step at inference time. Expected behavior: ROC AUC $\approx$ 0.93 and, at the default 0.5 threshold, ~79% precision / ~65% recall on data drawn from the same distribution as this training set; apply the tuned 0.40 threshold instead if recall matters more than precision for the campaign. Usage:

```
import joblib
pipeline = joblib.load('final_pipeline.pkl')
preds = pipeline.predict(X_new)
probs = pipeline.predict_proba(X_new)[:, 1]
preds_tuned = (probs >= 0.40).astype(int)
```

Caveat carried from Day 3: the underlying data shows a real gender income gap (the `sex_Female` feature was strongly negative in the Logistic Regression coefficients); this model should not be deployed for real targeting decisions without a fairness review, since it will reproduce that historical pattern. Library versions (scikit-learn 1.8.0, pandas 3.0.2, numpy 2.4.4, scipy 1.17.1, joblib 1.5.3) are pinned in the notebook's final cell for exact reproducibility.